

A falsification protocol for embodied spiking substrates

Separating substrate-carried information from readout-channel information

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Abstract

When a population of spiking neurons is read out to control a simulated body, “the substrate computes” is usually claimed from a decoded error or a mutual-information value. We show that such a claim is unfalsifiable unless the substrate is swapped for a *matched dead substrate* that preserves the single-neuron marginal firing statistics (mean, variance, silent fraction) while being statistically independent of the task, and unless the readout is itself controlled. We build a minimal embodied hub (2,000 neurons, fixed sparse wiring, one calibrated decision map) and run four gates: (B) substrate swap under a fixed readout; (C) readout-channel ablation (randomized projection into the decision); (D) cross-validated ridge readout of the *same* spikes; (E) memory-demanding tasks. The matched dead substrate is task-independent by construction; its empirical measures stay at the null floor under every readout ($MI \approx 0$, $TE = 0$, $\rho \approx 0$, all seeds, all profiles), and randomization leaves it at the same floor ($\Delta = 0.000$, 0/3 seeds). The living substrates do not uniformly survive: LIF carries the task through a cross-validated ridge readout ($\Delta RMSE_{\text{floor}} = +0.094$ vs. the mean-predictor floor; MI significant in 24/30 runs) whereas the apparent Izhikevich tracking seen under the fixed calibrated readout is largely a *readout-channel artifact*: randomizing the readout projection collapses *both* living substrates to the dead floor (3/3 seeds), and a cross-validated ridge readout of the same IZH spikes leaves RMSE essentially at the floor ($\Delta RMSE_{\text{floor}} = -0.008$). None of the substrates shows linearly decodable history in the present-window population state (Gate E negative), so memory claims are not certified here. We conclude: (i) matched dead-substrate controls for the *material* of a substrate are cheap, decisive, and should precede scoreboard comparisons; (ii) readout *robustness* is a stronger falsification test than a single scoreboard: task information that remains recoverable after randomizing and re-optimizing the readout is evidence carried by the substrate activity rather than by the calibrated interface alone; and (iii) open-loop diagnosis cannot certify memory, which must be demonstrated in the closed loop.

1 Introduction

Physical substrates for neural computation—in-vitro cultures [2], neuromorphic silicon [8, 14], or toy spiking hubs—are routinely evaluated by coupling them to a task and reporting a score: a decoded error, a correlation, a bit rate. Two distinct claims are usually conflated under that score. The first is *substrate-carried information*: spike patterns produced by the substrate contain task-relevant statistics. The second is *readout-channel information*: the fixed mapping used to translate spikes into a decision happens to recover (or inject) task statistics. A scoreboard alone cannot tell them apart, and the recent discussion around “intelligence in a dish” [2, 3, 4] shows that this conflation has consequences far beyond toy models.

The contribution of this note is a *falsification protocol* that separates the two. It rests on two ingredients, both cheap:

- **A matched dead substrate** (Gate B). The wiring, bias, drive and readout are kept fixed; only the per-neuron dynamics change. The dead substrate fires Poisson processes with per-neuron rates matched to the LIF marginal histogram, but is statistically independent of the task. Any substrate that cannot beat this null under a fixed readout is not falsifiably carrying the task.
- **Readout controls** (Gates C and D). Gate C randomizes the readout projection into the fixed decision map; if the apparent tracking collapses toward the dead floor, performance depends on the channel alignment. Gate D replaces the hand-tuned readout with a cross-validated ridge readout trained on the *same* spikes; task information that remains recoverable after *both* controls is evidence the substrate activity itself, rather than the calibrated interface alone, carries the signal. Throughout we reserve stronger language (“invariance”) for what the experiments cannot show, and use *readout robustness* for what they do.

Gate E then asks the harder question—whether present-window population activity carries task *history*—which is the minimum a memory claim needs. Throughout, the replication unit is the seed (6 seeds $\times$ 5 profiles for Gates B/D/E; 3 seeds for Gate C).

2 Related work

Five literatures touch the same question from different angles.

(1) Embodied neural computation. “Intelligence without representation” [5] and morphology-computation continua [6] established that behavior is jointly determined by body, wiring and control; more recent benchmarks formalize that for neuromorphic-body systems [7]. These works score systems, but do not isolate the substrate’s material from its interface.

(2) In-vitro substrates and the DishBrain debate. Kagan et al. reported task-relevant responses of cultured cortical neurons in a closed loop [2]; the follow-up correspondence [3, 4] turned precisely on whether the observed signal is attributable to the biology or to the measurement/readout interface. Our protocol is a minimal, fully deterministic enactment of that methodological critique.

(3) Spiking and reservoir computing. Spiking neuron models [8, 9, 10, 11] and liquid state machines / echo-state networks [12, 13, 14] show that random recurrent populations can support surprisingly capable online regression. The standard evaluation there is already readout-shaped (train a readout, report performance); what is usually missing is the *dead-material control*—the same readout trained on matched independent Poisson spikes [14].

(4) Population decoding and information measures. Neural coding theory provides the tools: population decoding with regularized linear readouts [23, 24, 25, 26], mutual information and its bias correction [15, 16, 17], and transfer entropy with surrogate-based significance [18, 19]. Our Gate D uses exactly this machinery; the dead-substrate null is the novel anchor against which those measures should be read.

(5) Null models and statistical rigor. Surrogate-data methods [20], permutation statistics [21], pseudo-replication warnings [22], and closed-loop validation practice in brain-machine interfaces [27, 28] all argue that a structurally matched null is the difference between an effect and an artifact. Gate C is a surrogate test applied to the *readout channel* rather than to the substrate.

3 Methods

3.1 Embodied hub and wiring

$N = 2,000$ neurons, $N_{\text{win}} = 200$ control windows of $C_W = 20$ integration steps (10 ms bins, 0.5 ms integration). The input projection $\mathbf{g} \in \mathbb{R}^{N \times 1}$ is standard-normal; recurrence $\mathbf{W}$ is sparse with connection probability $p_{\text{re}} = 0.01$ and entries drawn $\mathcal{N}(0, 1/\sqrt{Np_{\text{re}}})$ (variance-normalized so the network operates near stability); the readout projection is $\mathbf{w}_r = \mathbf{g}$ (channel aligned with the input projection), with a per-substrate scale k_{ro} (below). The decision map is the calibrated one of the reference system [1], $\hat{x} = \text{clip}((\theta_{\text{sig}} - 0.0615)/0.277, 0, 0.75)$, clamped to 0.75 m/s. Five control profiles (`baseline`, `multi_step`, `sine`, `descend`, `pulse`), six seeds $\{1, 7, 13, 29, 55, 91\}$.

3.2 Substrates and the matched dead null

Three dynamics on identical wiring, bias, drive and readout constants:

- **LIF:** leaky integrate-and-fire, $\tau = 20$ ms, threshold $v_{\text{th}} = 1$, reset 0, refractory 2 ms.
- **Izhikevich:** regular spiking parameters [“RS”, (0.02, 0.2, −65, 8)] integrated with the 0.5 ms step; adaptation dimension u and $a = 0.02$, $b = 0.2$, $c = -65$, $d = 8$.
- **Matched dead null (Poisson):** per-neuron events drawn independently each integration step from Bernoulli trials with probability r_i ; the r_i are estimated from LIF’s per-neuron firing in a constant-input ($u = 0.5$) calibration run [20], so the marginal per-neuron rates match. Table 1 reports the match quality explicitly; the null is *statistically independent* of u by construction.

The key design point is that the match is at the level of the *single-neuron marginal rate histogram*—mean μ_r , variance, and the fraction of near-silent neurons—not at the level of any temporal or population structure. This is what makes Gate B a *substrate* test rather than a firing-rate test. Table 1 (from `gate.b_summary.json`) gives $\mu_r = 8.09$ Hz, SD 15.30 Hz, Fano $\sigma^2/\mu = 28.9$ (overdispersed vs. a homogeneous Poisson) and 74% neurons with zero estimated rate.

Table 1: Match quality of the dead null to the LIF marginal: per-neuron mean rate, SD, Fano factor σ^2/μ , and the silent fraction (neuron with $\hat{r}_i \approx 0$). The null draws Bernoulli events at the matched per-neuron probabilities, independent of u .

statistic	LIF marginal (r_i)	matched IID Poisson	$N = 2000$ neurons	Fano
per-neuron mean rate (Hz)	8.091	8.091	both	–
per-neuron StdDev (Hz)	15.296	$\sqrt{\mu}$ only	–	28.916
fraction silent ($r_i = 0$)	0.740	$e^{-\mu}$ expected	1480/2000	–

3.3 Calibration without leakage

k_{ro} is fixed per substrate from a *constant-input* calibration run ($u = 0.5$, seed 7): $k_{\text{ro}} = 0.2/\langle \mathbf{w}_r \cdot \text{rate} \rangle$ sets the decision series at the middle of the map’s active range. Crucially, calibration never sees a *task-varying* input, so no task information leaks into the readout constant. The same k_{ro} survives Gate C: the randomized projection is rescaled to the same norm ($\|\mathbf{w}_r^{\text{rand}}\| = \|\mathbf{g}\|$), so the constant remains comparable. The Poisson null uses the LIF-derived k_{ro} plus a $k_{\text{ro}}^{\text{poisson}}$ from its own constant-input run, again $0.2/\langle \cdot \rangle$.

3.4 Gate C: readout-channel control

For each seed s , draw an independent zero-mean normal vector $\mathbf{w}_s^{\text{rand}}$, center it, and rescale to $\|\mathbf{g}\|$ (RNG seed 4242). Keep wiring, bias, drive and the fixed decision map; only the readout projection changes. If apparent tracking under the aligned channel survives randomization, it is substrate-independent; if it collapses to the dead floor, the apparent tracking depended on the readout alignment (whether the substrate activity *also* contains readout-recoverable task information is asked separately by Gate D).

3.5 Decoders (Gate D)

On the *same* recorded spike trains:

1. **canon**—the fixed calibrated channel $k_{\text{ro}} \mathbf{w}_r \cdot (\text{counts}/C_W) + \text{decide}$ (Gate B reference);
2. **ridge**—cross-validated L2-regularized linear readout of population counts with 13 lags on a log-grid in $[10^{-5}, 10]$, fit with the normal-equations/Woodbury solve on interleaved train/val windows (every phase appears in every fold) solved via the Woodbury identity and verified against a direct solve ($\max |\Delta| < 4 \times 10^{-10}$; `verify_ridge.py`);
3. **meanpd**—constant-predictor floor (var-around-run-mean);
4. **passthru**— $0.75 u$ directly (RMSE = 0 by construction, the no-substrate ceiling).

We report the *RMSE improvement over the mean-predictor floor*,

$$\Delta \text{RMSE}_{\text{floor}} = \text{RMSE}_{\text{meanpd}} - \text{RMSE}_{\text{ridge}},$$

taking the constant predictor as the zero-information baseline: positive values mean the spikes beat a constant. We deliberately avoid an information-unit name for $\Delta \text{RMSE}_{\text{floor}}$ because $-\text{RMSE}$ is a loss, not an entropy; the reader should not read it as a bit or nat quantity.

3.6 Information metrics

MI . Quantize command $\hat{x}$ and reference u to 12 equipopulated bins (percentile edges [2, 98]%), estimate the plug-in MI in nats, and bias-correct by 100 permutations of u [17]; report the two-sided empirical p . Runs with nominally zero-bias MI are declared “significant” when $p < 0.05$.

TE . Transfer entropy $\text{TE}(s \rightarrow x \mid x_{\text{past}})$ from the decoded-command event series s to the quantized command x , conditioning on the past of x [18], estimated by plug-in from the co-occurrence histogram and significance by the trial-shuffle / permutation baseline [19]. For a feed-forward hub, expected $\text{TE} \approx 0$; the dead null gives exactly 0 on every profile and seed.

Statistics and pseudo-replication. The unit of replication is the *seed* (wiring + initial state). Gate B/E pool 6 seeds; Gate D pools 6 seeds $\times$ 5 profiles = 30 runs. We report both per-seed outcomes (e.g. “significant MI in 24/30 runs”) and pooled means, and we flag seed-level counts rather than inflating n . This follows standard pseudo-replication guidance [22]: the 200 windows within a run are not independent evidence.

Experimental design at a glance.

Table 2: The four gates: what is changed, what is held fixed, and the question each gate answers.

Gate	Manipulation	Held fixed	Question
B (substrate swap)	dynamics $\rightarrow$ Poisson null with matched marginals	wiring, bias, drive, readout $\mathbf{w}_r = \mathbf{g}$, decision map	Does a task-independent dead substrate reproduce the score?
C (channel ablation)	readout projection $\rightarrow$ per-seed random $\mathbf{w}_r^{\text{rand}}$	wiring, drive, spikes, decision map	Does the apparent tracking survive breaking the aligned channel?
D (readout re-learn)	decoder $\rightarrow$ cross-validated ridge on the <i>same</i> spikes	spikes, task	Do the spike trains themselves contain readout-recoverable task information?
E (memory)	task $\rightarrow$ history-dependent (lag , integ , stair); decoder on present-window state	spikes, dynamics, readout	Is task <i>history</i> linearly decodable from the present population state?

Gates B–D decompose one score into (substrate material) vs. (interface); Gate E sets the bar any memory claim must clear. All four are run on the *same* hub, seeds and profiles.

4 Results

4.1 Gate B: substrate swap under a fixed readout

Table 3: Gate B: per-profile $\widehat{\text{RMSE}}$, MI (nats), ρ and TE-significance under the fixed calibrated readout (5 profiles, 6 seeds). “msd” is the across-seed mean.

profile	substrate	$\widehat{\text{RMSE}}$	MI	ρ	TE	p_{TE}
baseline	LIF	0.238	0.000	0.000	5e-04	0.253
baseline	Izhikevich	0.254	0.000	0.000	4e-05	0.400
baseline	Poisson (dead null)	0.399	0.000	0.000	0e+00	0.833
descend	LIF	0.167	1.034	0.933	0e+00	0.027
descend	Izhikevich	0.196	0.627	0.892	0e+00	0.000
descend	Poisson (dead null)	0.382	0.002	-0.006	0e+00	0.833
multi_step	LIF	0.131	0.681	0.994	0e+00	0.000
multi_step	Izhikevich	0.215	0.444	0.822	0e+00	0.000
multi_step	Poisson (dead null)	0.432	0.000	0.012	0e+00	0.833
pulse	LIF	0.124	0.574	0.958	0e+00	0.000
pulse	Izhikevich	0.294	0.241	0.570	0e+00	0.000
pulse	Poisson (dead null)	0.372	0.000	0.004	0e+00	0.833
sine	LIF	0.179	0.966	0.938	0e+00	0.027
sine	Izhikevich	0.236	0.614	0.879	0e+00	0.000
sine	Poisson (dead null)	0.375	0.000	-0.012	0e+00	0.833

Under the fixed calibrated readout, LIF and IZH both beat the matched dead null on every task-varying profile (Table 3); the dead null stays at $\text{MI} \approx 0$, $\text{TE} = 0$, $\rho \approx 0$. On **baseline** (constant input) *no* substrate carries information, as expected. Taken alone, Gate B would read as “the neural substrates compute”—this is precisely the reading that Gates C and D correct.

4.2 Gate C: does the channel carry it?

Table 4: Gate C: canonical RMSE under the aligned channel ($\mathbf{w}_r = \mathbf{g}$) vs. per-seed random readouts. “ d ” is $\text{RMSE}_{\text{random}} - \text{RMSE}_{\text{aligned}}$; “paired” is the fraction of seeds with $d > 0$. Both living substrates collapse (3/3 seeds); the matched dead null is unaffected.

profile	substrate	$\widehat{\text{RMSE}}_{\text{aligned}}$	$\widehat{\text{RMSE}}_{\text{random}}$	Δ	p_{paired}
sine	LIF	0.183	0.407	+0.224	3/3
sine	Izhikevich	0.235	0.396	+0.161	3/3
sine	Poisson (dead null)	0.407	0.407	+0.000	0/3
pulse	LIF	0.124	0.397	+0.273	3/3
pulse	Izhikevich	0.296	0.361	+0.065	3/3
pulse	Poisson (dead null)	0.397	0.397	+0.000	0/3

Randomizing the readout projection collapses *both* living substrates toward the dead floor: mean RMSE jumps from 0.12–0.24 (aligned) to 0.36–0.41 (randomized), in all three evaluated seeds per profile (Table 4). The matched dead null is unaffected ($\Delta = 0.000$, 0/3 seeds). Gate C therefore shows that performance under the fixed decision map *depends critically on the alignment of the readout projection* with the input projection. It does not by itself say the substrate is empty: a poor channel could also fail to expose genuinely present substrate information. Gate D settles that direction—whether the underlying spike trains retain readout-recoverable task information—by re-learning the readout on the same spikes.

4.3 Gate D: a readout-invariant answer

Table 5: Gate D: same spikes, decoder sweep (pooled 5 profiles $\times$ 6 seeds). $\Delta\text{RMSE}_{\text{floor}} = \text{RMSE}_{\text{meanpd}} - \text{RMSE}_{\text{decoder}}$ is the RMSE improvement over the mean-predictor (no-information) floor; MI in nats; MI_{sig} is the fraction of runs with significant MI.

substrate	decoder	$\widehat{\text{RMSE}}$	$\Delta\text{RMSE}_{\text{floor}}$	MI (nats)	ρ	MI_{sig}
LIF	fixed calibrated channel	0.168	-0.018	0.651	0.765	0.800
LIF	ridge (cross-validated)	0.055	0.094	0.391	0.733	0.800
Izhikevich	fixed calibrated channel	0.239	-0.090	0.385	0.633	0.800
Izhikevich	ridge (cross-validated)	0.157	-0.008	0.250	0.458	0.767
Poisson (dead null)	fixed calibrated channel	0.392	-0.243	0.000	-0.000	0.000
Poisson (dead null)	ridge (cross-validated)	0.158	-0.009	0.018	0.038	0.033

Table 5: a cross-validated ridge readout of *the same spikes* splits the substrates:

- **LIF**: RMSE drops to 0.055—the fixed channel had left information on the table (canon-to-ridge gap +0.113)—and the spikes genuinely carry the task: $\Delta\text{RMSE}_{\text{floor}} = +0.094$, MI significant in 24/30 runs.
- **Izhikevich**: the fixed-channel “tracking” does *not* persist: ridge collapses to 0.157, $\Delta\text{RMSE}_{\text{floor}} \approx 0$ (−0.008), MI significant in 23/30 runs but at about a quarter of the LIF rate. The Gate B IZH gap was a readout-channel artifact (confirmed by Gate C).
- **Dead null**: ridge 0.158 $\approx$ floor, MI ≈ 0 (frac. 0.03), TE = 0—the matched null remains at the floor under the cross-validated ridge readout [26, 25].

Recurrence ablation (nohub: w ignored) leaves canon RMSE unchanged to < 0.001 ; the tested recurrence contributes negligibly to the measured canonical RMSE, so none of the above is a recurrence artefact.

4.4 Gate E: is there memory?

Table 6: Gate E: RMSE improvement over the mean-predictor floor, $\Delta\text{RMSE}_{\text{floor}}$ (ridge minus floor), per profile, plus the input-only 8-tap FIR reference (positive *floor* values).

profile	LIF	Izhikevich	Poisson	input-FIR8
lag	-0.139	-0.055	-0.016	0.152
integ	-0.114	-0.104	-0.012	0.144
stair	-0.141	-0.025	0.010	-0.016

Negative result. Three history-dependent tasks (**lag**: 4-window delayed reference; **integ**: running integral over 8 windows; **stair**: staircase-level switch) decoded by the ridge on present-window spikes vs. an 8-tap FIR over the input only. $\Delta\text{RMSE}_{\text{floor}}$ is negative for every substrate (LIF -0.13 , IZH -0.06 , null -0.006), while the input-FIR solves the linearly recoverable ones ($+0.15$). Conclusion: no substrate shows *linearly decodable task history* in the present-window population state. Memory is not certified here; functional memory must be demonstrated in the closed loop, where previous activity feeds back into the input [27, 28].

4.5 Verification

Three independent checks confirm the results are not implementation artifacts:

- **verify_ridge.py**—Woodbury ridge equals a direct least-squares solve ($\max |\Delta| < 4 \times 10^{-10}$): Gate D numbers are the solver’s, not the optimization’s;
- **verify_null.py**—matched dead null on two out-of-family seeds {103, 201}: MI = 0, TE = 0, $\rho = 0$ exactly;
- **verify_wr.py**—per-seed independent random readouts ($\|\mathbf{w}^{\text{rand}}\| = \|\mathbf{g}\|$, RNG 4242): the randomization used in Gate C, verified against the aligned baseline and the null.

5 Discussion

5.1 The matched dead substrate is the falsification anchor

A scoreboard is necessary but not sufficient. Every number in Gate B that looks like “computation” is preceded by the two-line question: *does an independent Poisson process with the same marginal rates do it too?* When it does not, the material deserves a closer look; when it does, nothing follows about the substrate. The matched dead-null is the minimal falsification test because it fixes the only uncontrolled variable—the dynamics—while wiring, readout and decision stay put. It is cheap (a rate estimate plus a Poisson generator) and it should precede scoreboard comparisons in the embodied-AI literature [2, 3, 14].

5.2 Readout robustness is the substantive test

Gate C makes the channel visible by breaking it: both living substrates collapse when $\mathbf{w}_r$ stops matching $\mathbf{g}$. Gate D then asks whether a readout *trained* on the same spikes can recover the task: LIF survives ($\Delta\text{RMSE}_{\text{floor}} = +0.094$), IZH does not ($\Delta\text{RMSE}_{\text{floor}} = -0.008$). Readout choice is therefore a substantive modeling decision, not a detail. Claims like “this material computes ϕ ” must be re-read as “this material’s spike statistics, *via* some channel, contain readout-recoverable task-relevant structure,” and the channel must be shown to be redundant—or the claim collapses to a statement about the interface, exactly the misreading flagged in the DishBrain correspondence [3].

5.3 What we are not claiming

The measured canonical RMSE is unaffected by dropping the recurrent connections (nohub ablation, Gate D); the tested recurrence contributes negligibly to the results, and Gate E is negative, so no statement here concerns memory. None of the numbers is a closed-loop result. A closed-loop falsification battery is left to future work (§5.5); this note’s protocol is the *open-loop instrument* that any closed-loop claim should first pass. Only LIF of the three materials survives the readout controls as a carrier of task information, and only for the rate-code tasks.

5.4 Limits

(i) The ridge readout is linear; a nonlinear decoder could in principle recover task information we miss (we deliberately test the weakest nontrivial decoder). (ii) The matched marginal is *single-neuron*: a null with matched population statistics (e.g. correlations) would be a stronger version. (iii) Gate C randomizes one projection; joint randomization of wiring and channel is a still stronger ablation. (iv) The seeds (6) bound the exact paired permutation resolution at 1/64; per-seed reporting keeps the pseudo-replication honest [22].

5.5 Future work

Three directions follow directly: (a) nonlinear decoders (MLP) against the same matched null to bound how much linearity hides; (b) matched-dead-null with population-level statistics (pairwise correlations) to test substrate *structure* rather than rates; (c) replaying the *real* closed-loop data through the same four gates, converting this open-loop instrument into a closed-loop falsification battery [27, 1].

6 Reproducibility and data

- Code: `sandbox/` git repo; Gates B/D/E runners (`gate_b/run.py`, `gate_d/run.py`, `gate_e/run.py`) print JSON summaries into `results/`; tables/figures are *generated* from those JSON artifacts by `build_figures_tables.py` (no hand-typed numbers).
- Gate C: `verify/verify_wr.py` (per-seed random readouts, RNG 4242) writes `verify_wr_summary.json`.
- Environment: Python 3.12, numpy 2.5, matplotlib; venv at `02\_cl/.venv312/`; `requirements.txt` pins versions.
- Scope: this note’s claims rest on Gates B/C/D/E only (all artifacts above). A companion closed-loop battery lives in the separate `surrogate_cl/` repository; it is *not* regenerated here and none of the results in this note depend on it [1].
- Video: `video/pulse_raster.mp4`, `video/g1_walk.mp4`, `video/g1_closed_loop.mp4`; produced by `make_video.py`, `robot_video.py`, `closed_loop_video.py`.
- Build note: this PDF compiled with MiKTeX (`latexmk -pdf`); no Overleaf-only packages.

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